MTH636 – Game Theory

Lecture 19: Normal Form equivalence & SPE

1 Normal Form Representation of Extensive Form Games

Suppose Γ is an extensive form game (potentially with chance nodes). We can always translate it into an equivalent normal form game:

$$\Gamma = \langle N, V, E, n_0, (V_i)_{i\in N\cup\{0\}}, (p_n)_{n\in V_0}, (u_i)_{i\in N}\rangle$$

The equivalent normal form game is given by $\langle N, (S_i)_{i\in N}, (\hat{u}_i)_{i\in N}\rangle$, where:

- N is the same set of players.
- S_i is the set of pure strategies for player i derived from the extensive form game (recall $s_i : V_i \to \cup A(n)$).
- $\hat{u}_i(s)$ is the expected utility of the strategy profile $s = (s_1, \ldots, s_n)$.

1.1 Utility Computation

If the game has **no chance nodes**, a strategy profile s uniquely traces a path to exactly one leaf node π. In this case:

$$\hat{u}_i(s) = u_i(\pi)$$

If the game **has chance nodes**, a strategy profile s no longer leads to a single leaf. Instead, because Nature introduces randomness at V_0 nodes, the profile s induces a probability distribution over several possible leaf nodes. In this case, utilities are calculated as the **expected** utility according to the probability distribution over the leaf nodes π. For example, if player 1 plays L and player 2 plays T, and a chance node assigns 50/50 probability to reaching leaf n_3 or n_4, then:

$$\hat{u}_1(L, T) = \frac{1}{2}u_1(n_3) + \frac{1}{2}u_1(n_4)$$

Note on Zermelo's Theorem: When chance nodes are introduced, Zermelo/Von Neumann's theorem (which guaranteed a pure winning or drawing strategy) generally *does not hold*. This is because the introduction of expected utilities smooths the discrete $\{1, -1, 0\}$ outcomes into fractional expected payoffs, breaking the deterministic Win/Loss/Draw structure.

2 Nash Equilibrium vs Subgame Perfect Equilibrium

Once translated into normal form, we can find the Nash Equilibria using the standard definition:

Definition: Nash Equilibrium

A strategy profile s^* is a Nash Equilibrium if:

$$u_i(s^*) \geq u_i(s_i, s^*_{-i}) \quad \forall i \in N, \ \forall s_i \in S_i$$

2.1 The Problem with Normal Form NE

Consider the Sequential Battle of the Sexes from Lecture 15:

- Man chooses F or D first.
- Woman observes and chooses F or D.

The normal form matrix has a Nash Equilibrium at (D, DD) which yields payoffs $(1, 2)$.

However, let's analyze the strategy DD ("I will play Dance if you play Football, and I will play Dance if you play Dance"). If the Man actually chose Football (F), the game moves to the subgame where the Woman must choose. Her payoffs are $(2, 1)$ for choosing F and $(0, 0)$ for choosing D. Clearly, in that subgame, she would rationally choose F to get 1 instead of 0.

Thus, her strategy to play D in response to F is a **non-credible threat**. She is threatening to tank both of their payoffs to $(0, 0)$ to scare the Man into playing D initially. While (D, DD) is mathematically a Nash Equilibrium of the whole game, it relies on irrational behavior in a specific subgame.

2.2 Subgame Perfect Equilibrium (SPE)

To eliminate non-credible threats, we refine our equilibrium concept for extensive form games.

Definition: Subgame Perfect Equilibrium (SPE)

A strategy profile is a Subgame Perfect Equilibrium (SPE) if it constitutes a Nash Equilibrium in **every** subgame of the original game.

By demanding that players act rationally not just overall, but from *every single decision node* (even ones off the equilibrium path), SPE effectively rules out empty threats. In the example above, (D, DD) is an NE, but **not** an SPE. The profile (F, FD) is both an NE and an SPE.